Supplementary Materials for "Thermal biology of mosquito-borne disease"

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Supplementary Tables (Tables S1-S2)

Table S1. Published species distribution models for vectors and vector-borne diseases.

Table S2. Trait thermal response functions fit across systems.

Supplementary Figures (Figures S1-S4)

Fig. S1. Validation of our malaria temperature-dependent R_0 model (Mordecai *et al.* 2013) in comparison with a previous model (Parham & Michael 2010).

Fig. S2. Validation of our dengue, chikungunya, and Zika virus temperature-dependent R_0 model in *Aedes aegypti* (Mordecai *et al.* 2017).

Fig. S3. Validation of our West Nile virus temperature-dependent R_0 model (Shocket et al., in prep).

Fig. S4. Validation of our Ross River virus temperature-dependent R_0 model (Shocket *et al.* 2018).

Table S1. Trait thermal response functional forms, parameters (including thermal limits), and optima. Medians and 95% CIs. T_{min}, T_{max}, and T_{opt} in degrees Celcius. F(x) is the functional form of the thermal response: B = Brière ($qT[T - T_{min}][T_{max} - T]^{1/2}$), Q = quadratic ($-q[T - T_{min}][T - T_{max}]$), Q* = alternative quadratic ($qT^2 - rT + s$), and L = linear ($-mT + z$). Traits with functional forms Q* and L are listed separately below.

Trait / System	***F(x)***	***q* (CIs)**	***T_{min}* (CIs)**	***T_{max}* (CIs)**	***T_{opt}* (CIs)**
Biting rate (*a*)					
Ae. aegypti	B	$2.02 \cdot 10^{-4}$ (1.26–$2.88 \cdot 10^{-4}$)	13.8 (7.7–17.1)	40.0 (40.0–40.4)	33.8 (33.0–34.3)
Ae. albopictus	B	$1.90 \cdot 10^{-4}$ (1.33–$2.68 \cdot 10^{-4}$)	10.4 (5.4–14.4)	38.1 (36.8–40.9)	31.8 (30.8–33.7)
An. pseudopunctipennis	B	$3.43 \cdot 10^{-4}$ (1.28–$8.21 \cdot 10^{-4}$)	17.1 (2.6–23.3)	38.7 (35.6–44.2)	33.2 (30.4–37.3)
Cx. annulirostris	B	$1.23 \cdot 10^{-4}$ (1.02–$1.70 \cdot 10^{-4}$)	3.8 (0.7–8.5)	39.1 (38.0–40.0)	31.8 (30.8–32.6)
Cx. pipiens	B	$1.70 \cdot 10^{-4}$ (1.18–$2.29 \cdot 10^{-4}$)	9.4 (2.8–13.4)	39.6 (37.9–40.6)	32.7 (31.3–33.6)
Cx. quinquefasciatus	B	$7.28 \cdot 10^{-5}$ (5.31–$11.8 \cdot 10^{-5}$)	3.1 (0.1–10.9)	39.3 (38.0–40.8)	31.9 (30.6–33.3)
Cx. tarsalis	B	$1.67 \cdot 10^{-4}$ (0.87–$2.56 \cdot 10^{-4}$)	2.3 (0.1–9.4)	32.0 (30.6–41.7)	25.9 (24.8–33.9)
Cs. melanura	B	$1.87 \cdot 10^{-4}$ (1.49–$2.31 \cdot 10^{-4}$)	7.8 (5.5–11.4)	31.8 (31.0–33.4)	26.4 (25.7–27.9)
Fecundity					
Ae. aegypti (*EFD*)	B	$8.16 \cdot 10^{-3}$ (4.32–$15.3 \cdot 10^{-3}$)	14.7 (7.7–20.4)	34.4 (34.0–36.2)	29.6 (28.5–31.1)
Ae. albopictus (*TFD*)	B	$4.77 \cdot 10^{-2}$ (3.43–$7.01 \cdot 10^{-2}$)	7.9 (3.4–13.3)	35.6 (35.1–36.9)	29.4 (28.7–30.6)
Cx. annulirostris (*EFD*)	Q	$5.65 \cdot 10^{-3}$ (3.90–$8.71 \cdot 10^{-3}$)	14.8 (11.9–17.2)	31.3 (30.3–32.9)	27.0 (26.2–28.4)
Cx. annulirostris (*nLR*)	Q	$2.75 \cdot 10^{-3}$ (1.99–$3.29 \cdot 10^{-3}$)	17.1 (15.6–17.9)	37.1 (36.4–38.4)	27.2 (26.6–27.6)
Cx. pipiens (*EFGC*)	Q	$5.98 \cdot 10^{-1}$ (4.31–$7.91 \cdot 10^{-1}$)	5.3 (2.6–8.5)	38.9 (36.2–41.8)	22.1 (20.1–24.4)
Cx. quinquefasciatus (*ER*)	Q	$6.36 \cdot 10^{-1}$ (4.50–$9.05 \cdot 10^{-1}$)	5.0 (1.3–9.8)	37.7 (34.8–40.7)	21.4 (18.9–24.4)
Proportion ovipositing (*pO*)					
Cx. pipiens	Q	$4.45 \cdot 10^{-3}$ (2.54–$7.77 \cdot 10^{-3}$)	8.2 (4.6–12.1)	33.2 (30.1–37.5)	20.8 (18.6–23.4)
Cx. quinquefasciatus	B	$6.67 \cdot 10^{-4}$ (5.80–$7.91 \cdot 10^{-4}$)	1.7 (0.2–4.8)	31.8 (31.1–32.2)	24.9 (21.8–26.0)
Cs. melanura	Q	$6.31 \cdot 10^{-3}$ (4.52–$7.89 \cdot 10^{-3}$)	8.7 (6.9–10.4)	33.6 (32.5–35.4)	20.7 (16.9–22.3)
Mosquito Dev. Rate (*MDR*)					
Ae. aegypti	B	$7.83 \cdot 10^{-5}$ (5.80–$10.0 \cdot 10^{-5}$)	11.6 (6.9–14.7)	39.1 (39.0–39.7)	32.7 (32.1–33.3)
Ae. albopictus	B	$6.33 \cdot 10^{-5}$ (4.74–$8.37 \cdot 10^{-5}$)	8.7 (4.3–12.2)	39.6 (37.9–41.9)	32.6 (31.4–34.4)
Ae. camptorhynchus	B	$4.29 \cdot 10^{-5}$ (2.50–$8.29 \cdot 10^{-5}$)	9.3 (1.8–19.1)	38.8 (37.2–40.3)	32.2 (30.6–34.0)
Ae. notoscriptus	B	$6.81 \cdot 10^{-5}$ (5.43–$8.56 \cdot 10^{-5}$)	9.6 (6.4–12.9)	38.8 (37.1–40.0)	32.2 (30.8–33.4)
Ae. triseriatus	B	$4.30 \cdot 10^{-5}$ (3.01–$5.83 \cdot 10^{-5}$)	0.8 (0–7.5)	36.5 (34.6–39.5)	29.3 (27.8–31.9)
Ae. vexans	B	$4.33 \cdot 10^{-5}$ (3.34–$5.50 \cdot 10^{-5}$)	1.9 (0.1–10.5)	38.2 (37.0–39.5)	30.9 (29.8–32.2)

An. gambiae	B	$9.52 \cdot 10^{-5}$ (4.41–$19.6 \cdot 10^{-5}$)	15.4 (10.0–19.9)	34.7 (33.0–38.3)	29.9 (28.4–32.7)
Cx. annulirostris	B	$6.82 \cdot 10^{-5}$ (5.47–$8.47 \cdot 10^{-5}$)	11.1 (7.7–14.4)	39.2 (37.9–40.1)	32.8 (31.6–33.6)
Cx. pipiens	B	$3.76 \cdot 10^{-5}$ (3.36–$4.47 \cdot 10^{-5}$)	0.1 (0–4.0)	38.5 (37.6–39.8)	30.9 (30.2–31.9)
Cx. quinquefasciatus	B	$4.14 \cdot 10^{-5}$ (3.46–$5.26 \cdot 10^{-5}$)	0.1 (0–5.5)	38.6 (37.4–40.6)	31.0 (30.0–32.6)
Cx. tarsalis	B	$4.12 \cdot 10^{-5}$ (3.15–$5.47 \cdot 10^{-5}$)	4.3 (0–8.4)	39.9 (37.9–42.2)	32.3 (31.0–34.0)
Cs. melanura	B	$2.74 \cdot 10^{-5}$ (1.64–$4.72 \cdot 10^{-5}$)	8.6 (0–16.8)	37.6 (35.1–40.4)	31.1 (28.7–33.7)
Immature survival					
Ae. aegypti (p_{EA})	Q	$5.99 \cdot 10^{-3}$ (5.16–$6.85 \cdot 10^{-3}$)	13.6 (12.5–14.5)	38.3 (37.6–39.0)	25.9 (25.4–26.4)
Ae. albopictus (p_{EA})	Q	$3.56 \cdot 10^{-3}$ (2.65–$4.84 \cdot 10^{-3}$)	9.1 (6.3–11.6)	39.3 (37.2–41.6)	24.2 (22.6–25.7)
Ae. camptorhynchus (p_{LA})	Q	$2.88 \cdot 10^{-3}$ (1.99–$4.83 \cdot 10^{-3}$)	5.3 (1.9–10.1)	38.6 (35.7–42.3)	22.0 (20.2–24.6)
Ae. notoscriptus (p_{LA})	Q	$5.68 \cdot 10^{-3}$ (4.37–$7.32 \cdot 10^{-3}$)	9.1 (7.1–11.0)	36.2 (34.7–37.9)	20.0 (17.0–23.6)
Ae. triseriatus (p_{LA})	Q	$3.26 \cdot 10^{-3}$ (1.95–$5.18 \cdot 10^{-3}$)	8.3 (4.9–11.4)	35.7 (32.9–39.7)	22.0 (19.9–24.6)
Ae. vexans (p_{LA})	Q	$3.29 \cdot 10^{-3}$ (2.65–$4.24 \cdot 10^{-3}$)	9.1 (8.1–10.6)	40.8 (38.4–43.6)	25.0 (23.9–26.2)
An. gambiae (p_{EA})	Q	$7.65 \cdot 10^{-3}$ (4.20–$12.5 \cdot 10^{-3}$)	14.5 (11.4–16.4)	34.1 (32.1–36.6)	24.3 (22.9–25.5)
Cx. annulirostris (p_{LA})	Q	$5.35 \cdot 10^{-3}$ (4.31–$6.65 \cdot 10^{-3}$)	14.9 (13.3–16.5)	39.1 (37.9–40.4)	27.0 (26.0–28.0)
Cx. pipiens (p_{LA})	Q	$3.60 \cdot 10^{-3}$ (2.96–$4.42 \cdot 10^{-3}$)	7.8 (6.1–9.3)	38.4 (37.1–39.9)	23.1 (22.2–24.0)
Cx. quinquefasciatus (p_{LA})	Q	$4.26 \cdot 10^{-3}$ (3.51–$5.17 \cdot 10^{-3}$)	8.9 (7.6–9.9)	37.7 (36.2–39.2)	23.3 (22.5–24.0)
Cx. tarsalis (p_{LA})	Q	$2.12 \cdot 10^{-3}$ (1.52–$3.08 \cdot 10^{-3}$)	5.9 (3.0–8.8)	43.1 (39.8–47.5)	24.6 (22.9–26.4)
Cs. melanura (p_{LA})	Q	$3.03 \cdot 10^{-3}$ (1.55–$5.68 \cdot 10^{-3}$)	10.1 (5.7–15.1)	36.2 (32.8–40.7)	23.2 (20.4–26.5)
Egg viability (*EV*)					
Cx. annulirostris	Q	$6.44 \cdot 10^{-3}$ (3.37–$12.7 \cdot 10^{-3}$)	14.5 (10.6–17.2)	38.0 (33.7–42.0)	26.2 (22.6–27.8)
Cx. pipiens	Q	$2.11 \cdot 10^{-3}$ (1.36–$3.05 \cdot 10^{-3}$)	3.2 (0.5–7.1)	42.6 (39.7–48.3)	23.0 (20.7–26.3)
Cx. quinquefasciatus	B	$0.47 \cdot 10^{-3}$ (0.34–$0.62 \cdot 10^{-3}$)	13.6 (9.3–16.8)	38.0 (37.2–38.7)	32.1 (31.3–32.7)
Cx. theileri	Q	$2.54 \cdot 10^{-3}$ (1.86–$3.41 \cdot 10^{-3}$)	5.5 (2.6–8)	45.4 (42.4–49.0)	23.6 (18.2–27.0)
Lifespan (*lf*)					
Ae. aegypti–DENV	Q	$1.44 \cdot 10^{-1}$ (1.05–$2.21 \cdot 10^{-1}$)	9.0 (6.7–12.5)	37.7 (35.8–40.0)	23.4 (22.2–24.8)
Ae. aegypti–ZIKV	Q	$2.96 \cdot 10^{-1}$ (1.48–$4.89 \cdot 10^{-1}$)	11.7 (5.1–14.5)	37.2 (34.9–39.7)	24.4 (21.7–25.8)
Ae. albopictus	Q	$13.9 \cdot 10^{-1}$ (7.73–$23.1 \cdot 10^{-1}$)	13.5 (10.4–16)	31.4 (29.5–34.0)	22.5 (21.1–23.7)
Cx. annulirostris	Q	$2.42 \cdot 10^{-1}$ (1.77–$3.04 \cdot 10^{-1}$)	13.1 (10.6–14.6)	33.6 (33.0–34.5)	23.4 (22.4–24.0)
Transmission probability (*b*)					
DENV | *Ae. aegypti*	B	$8.33 \cdot 10^{-4}$ (5.42–$12.4 \cdot 10^{-4}$)	17.2 (12.1–20.9)	35.8 (35.1–36.8)	31.0 (29.9–32.1)
DENV | *Ae. albopictus*	B	$7.23 \cdot 10^{-4}$ (4.62–$10.7 \cdot 10^{-4}$)	16.0 (11.3–19.8)	36.3 (36.0–37.1)	31.2 (29.9–32.1)
ZIKV | *Ae. aegypti*	B	$3.44 \cdot 10^{-3}$ (1.83–$5.65 \cdot 10^{-3}$)	22.9 (20.5–23.9)	38.4 (36.6–40.5)	30.6 (29.6–31.5)

SLEV | *Cx. tarsalis*	Q	$2.98 \cdot 10^{-3}$ (1.63–$5.31 \cdot 10^{-3}$)	10.8 (6.2–14.2)	41.6 (36.8–49.1)	26.2 (23.5–29.7)
WEEV | *Cx. tarsalis*	Q	$3.17 \cdot 10^{-3}$ (1.65–$5.06 \cdot 10^{-3}$)	8.2 (5.1–10.7)	33.5 (31.0–38.9)	20.9 (19.2–23.2)
WNV | *Cx. tarsalis*	Q	$2.94 \cdot 10^{-3}$ (1.91–$4.48 \cdot 10^{-3}$)	11.3 (7.6–14.0)	41.9 (37.7–47.0)	26.6 (23.9–29.3)
Infection probability (*c*)					
DENV | *Ae. aegypti*	B	$4.88 \cdot 10^{-4}$ (3.43–$6.53 \cdot 10^{-4}$)	12.7 (4.3–17.1)	37.4 (35.8–39.5)	31.5 (29.6–33.5)
DENV | *Ae. albopictus*	B	$4.33 \cdot 10^{-4}$ (3.37–$5.78 \cdot 10^{-4}$)	2.8 (0–11.2)	36.7 (36.1–38.2)	29.6 (24.7–31.3)
SINV | *Ae. taeniorhynchus*	Q	$1.24 \cdot 10^{-3}$ (0.75–$2.17 \cdot 10^{-3}$)	1.4 (0–9.1)	48.4 (40.8–57.1)	25.4 (21.0–31.1)
SINV | *Cx. pipiens*	Q	$1.33 \cdot 10^{-3}$ (0.47–$2.30 \cdot 10^{-3}$)	0 (0–0)	35.0 (28.1–61.1)	17.5 (14.1–30.5)
WNV | *Cx. pipiens*	Q	$2.56 \cdot 10^{-3}$ (2.05–$3.19 \cdot 10^{-3}$)	15.6 (14.3–16.6)	52.2 (48.4–56.6)	33.9 (31.9–36.1)
SLEV | *Cx. tarsalis*	Q	$2.03 \cdot 10^{-3}$ (1.28–$3.07 \cdot 10^{-3}$)	8.8 (6.6–10.6)	43.7 (38.9–51.4)	26.2 (24.2–29.7)
WEEV | *Cx. tarsalis*	Q	$3.04 \cdot 10^{-3}$ (2.52–$3.68 \cdot 10^{-3}$)	1.3 (0.4–2.9)	38.8 (36.7–41.5)	15.5 (13.4–19.7)
Vector competence (*bc*)					
RVFV | *Ae. taeniorhynchus*	Q	$1.51 \cdot 10^{-3}$ (1.03–$2.05 \cdot 10^{-3}$)	7.1 (2.8–9.8)	42.3 (39.3–46.5)	24.7 (22.0–27.0)
EEEV | *Ae. triseriatus*	Q	$1.51 \cdot 10^{-3}$ (0.96–$2.24 \cdot 10^{-3}$)	7.0 (2.9–11.9)	50.3 (42.3–63.1)	28.8 (23.6–35.8)
P. fal.| *An. quadrimaculatus*	Q	$6.13 \cdot 10^{-3}$ (4.85–$7.63 \cdot 10^{-3}$)	12.6 (11.2–13.8)	35.9 (34.4–37.6)	24.3 (23.3–25.2)
RRV | *Cx. annulirostris*	Q	$2.78 \cdot 10^{-3}$ (2.02–$4.27 \cdot 10^{-3}$)	6.1 (1.6–9.6)	42.8 (39.6–44.9)	24.4 (22.4–25.6)
WNV | *Cx. pipiens*	Q	$3.05 \cdot 10^{-3}$ (1.68–$4.87 \cdot 10^{-3}$)	16.8 (15–17.9)	38.9 (36.1–44.1)	27.8 (26.6–30.1)
WEEV | *Cx. tarsalis*	Q	$1.17 \cdot 10^{-3}$ (0.55–$2.36 \cdot 10^{-3}$)	5.1 (0.6–13.3)	37.0 (33.5–46.0)	21.4 (18.1–27.3)
WNV | *Cx. univittatus*	Q	$2.32 \cdot 10^{-3}$ (1.58–$3.68 \cdot 10^{-3}$)	4.2 (1.5–7.1)	45.2 (39.6–53.0)	23.7 (19.4–27.3)
Parasite Dev. Rate (*PDR*)					
DENV | *Ae. aegypti*	B	$6.13 \cdot 10^{-5}$ (3.98–$11.9 \cdot 10^{-5}$)	10.3 (3.7–18.2)	45.6 (40.1–53.5)	37.7 (33.8–43.8)
ZIKV | *Ae. aegypti*	B	$1.68 \cdot 10^{-4}$ (0.62–$3.21 \cdot 10^{-4}$)	19.7 (4.9–23.7)	42.5 (38.8–44.9)	36.5 (33.2–38.9)
RVFV | *Ae. taeniorhynchus*	B	$8.84 \cdot 10^{-5}$ (2.51–$15.5 \cdot 10^{-5}$)	9.0 (5.4–13.8)	45.9 (41.9–50.3)	37.8 (34.5–41.3)
EEEV | *Ae. triseriatus*	B	$7.05 \cdot 10^{-5}$ (5.21–$9.68 \cdot 10^{-5}$)	11.6 (7.0–16.4)	44.8 (40.6–49.4)	37.2 (33.8–41.1)
P. fal.| *Anopheles* spp.	B	$1.06 \cdot 10^{-4}$ (0.88–$1.26 \cdot 10^{-4}$)	14.1 (13.1–15.2)	34.2 (33.5–34.8)	29.2 (28.7–29.7)
MVEV | *Cx. annulirostris*	B	$1.01 \cdot 10^{-4}$ (0.64–$1.55 \cdot 10^{-4}$)	12.8 (6.6–19.0)	41.9 (38.6–45.0)	35.2 (32.2–37.8)
RRV | *Cx. annulirostris*	B	$1.34 \cdot 10^{-4}$ (0.86–$2.15 \cdot 10^{-4}$)	5.9 (1.3–13.0)	41.3 (36.1–45.0)	33.6 (29.6–37.0)
WNV | *Cx. pipiens*	B	$7.38 \cdot 10^{-5}$ (5.38–$9.94 \cdot 10^{-5}$)	11.4 (7.3–15.0)	45.2 (40.7–50.3)	37.5 (33.8–41.6)
WNV | *Cx. quinquefasciatus*	B	$7.12 \cdot 10^{-5}$ (4.58–$10.2 \cdot 10^{-5}$)	19.0 (12.9–21.0)	44.1 (38.8–50.4)	37.7 (33.6–42.7)
SLEV | *Cx. tarsalis*	B	$7.11 \cdot 10^{-5}$ (5.60–$8.95 \cdot 10^{-5}$)	12.8 (10.3–14.3)	45.2 (40.2–51.5)	37.7 (33.8–42.6)
WEEV | *Cx. tarsalis*	B	$6.43 \cdot 10^{-5}$ (4.44–$10.4 \cdot 10^{-5}$)	4.0 (0–12.6)	44 .0(38.3–50.9)	35.7 (31.0–41.4)
WNV | *Cx. tarsalis*	B	$6.57 \cdot 10^{-5}$ (5.11–$8.85 \cdot 10^{-5}$)	11.2 (7.9–14.9)	44.7 (40.4–49.4)	37.0 (33.6–40.9)
WNV | *Cx. univittatus*	B	$7.54 \cdot 10^{-5}$ (4.13–$11.1 \cdot 10^{-5}$)	10.2 (7.1–15.3)	34.4 (31.2–51.1)	28.8 (26.1–42.5)

Trait / System	***F(x)***	***q* (CIs)**	***r* (CIs)**	***s* (CIs)**
Mortality rate (*μ*)				
An. gambiae	Q*	$1.10 \cdot 10^{-3}$ ($9.29 \cdot 10^{-4}$ – $1.27 \cdot 10^{-3}$)	$4.90 \cdot 10^{-2}$ ($4.10 \cdot 10^{-2}$ – $5.66 \cdot 10^{-2}$)	0.609 (0.527 – 0.682)

Trait / System	***F(x)***	**m (CIs)**	***z* (CIs)**	***Tmax* = *z/m* (CIs)**
Lifespan (*lf*)				
Ae. taeniorhynchus	L	2.02 (1.59–3.19)	85.9 (73.8–117.6)	42.7 (34.5–48.5)
Cx. pipiens	L	4.86 (3.83–5.84)	169.8 (142.1–195.6)	34.9 (32.9–37.9)
Cx. quinquefasciatus	L	3.80 (1.85–5.29)	136.3 (86.8–174.0)	35.9 (32.1–48.5)
Cx. tarsalis	L	1.69 (1.12–2.24)	69.6 (55.8–83.5)	41.3 (36.6–50.8)

Table S2. Published species distribution models for vectors and vector-borne diseases. Columns indicate the study, pathogen species (where applicable), vector species, vector life stage (where applicable), the temperature measurement(s) included in the model, and the reported relationship between temperature and occurrence probability.

Reference	Disease	Vector/Host	Life stage	Model	Temperature measures	Relationship reported
(Bishop *et al.* 2000)	Bluetongue	*Culicoides brevitarsis*	Adult	LOESS	Weekly or monthly mean temperature	Time from occurrence and distance from site
(Calvete *et al.* 2009)	Bluetongue	Animal	NA	GLM	Annual mean temperature	Parameters
(Cappai *et al.* 2018)	Bluetongue	Sheep, goat	NA	Logistic mixed model	Decadal mean temperature	Odds ratios
(Conte *et al.* 2004)	Bluetongue	*Cu. imicola*	Adult	Spatial process model	Daily minimum temperature	Not given
(Gao *et al.* 2017)	Bluetongue	Sheep	NA	MaxEnt	Monthly mean temperature, monthly max temperature, monthly min temperature, monthly temperature range	Not given
(Mukhopadhyay *et al.* 2017)	Bluetongue	*Cu. orientalis*	Adult	MaxEnt	Bioclim (annual mean temperature, mean temperature of coldest quarter, mean temperature of warmest quarterly)	Not given
(Purse *et al.* 2004)	Bluetongue	*Cu. pulicaris, Cu. newsteadi, Cu. obsoletus, Cu. imicola*	Adult	Stepwise discriminant analysis	Mean, variance, and minimum land surface temperature, minimum air temperature	Not given
(Rigot *et al.* 2012)	Bluetongue	*Cu. imicola*	Adult	GLM	Daily minimum and maximum temp, monthly mean, minimum, and maximum land surface temperature	Parameters
(Tatem *et al.* 2003)	Bluetongue	*Cu. imicola*	Adult	Stepwise discriminant analysis	Monthly maximum land surface temperature, monthly mean air temperature, monthly variance land surface temperature	Not given
(Escobar *et al.* 2016a)	CHIKV	*Aedes aegypti, Aedes albopictus*	NA	NicheA	WorldClim (annual mean temperature, mean diurnal temperature range, isothermality, temperature seasonality, maximum temperature of the warmest month, minimum temperature of the coldest month, temperature annual range,	Central values of climate conditions studied were around 25°C, the temperature corresponding to high reported CHIKV R_o

					mean temperature of warmest quarter, mean temperature of coldest quarter)	
(Nsoesie *et al.* 2016)	CHIKV	*Aedes aegypti, Aedes albopictus*	NA	Boosted regression trees	Land surface temperature	Not given
(Bhatt *et al.* 2013)	DENV	*Aedes aegypti*	NA	Boosted regression trees	Monthly minimum, maximum, and mean temperature restricted by DENV transmission suitability	Not given
(Mweya *et al.* 2016)	DENV	*Aedes aegypti*	Larvae, pupae, adults	MaxEnt	Bioclim (annual temperature range, mean temperature of warmest quarter, temperature seasonality, minimum temperature of coldest month, maximum temperature of warmest month, mean diurnal range)	Not given
(Alegana *et al.* 2014)	Malaria	Human	NA	Bayesian conditional autoregressive model	Temperature suitability index	Parameters
(Foley *et al.* 2014)	Malaria	*Anopheles albitarsis, An. deaneorum, An. janconnae, An. marajoara, An. oryzalimnetes, An. albitarsis F, An. albitarsis G, An. albitarsis I, An. albitarsis*	Adult	MaxEnt	Bioclim (mean diurnal range, minimum temperature of coldest month, annual temperature range, mean temperature of wettest quarter, mean temperature of driest quarter, mean temperature of warmest quarter, mean temperature of coldest quarter)	Not given
(Hundessa *et al.* 2018)	Malaria	Human	NA	GAM	Annual mean temperature	Effect size
(Arsevska *et al.* 2016)	RVF	*Aedes vexans, Anopheles multicolor, Culex perexiguus,*	NA	Multiple criteria decision making with weighted linear combination	Monthly mean temperature	Description and range of function
(Clements *et al.* 2006)	RVF	Human, animal	NA	Multiple criteria decision making with weighted linear combination	Land surface temperature	Description and range of function
(Clements *et al.* 2007)	RVF	Cattle, goat, sheep	NA	Bayesian spatial model	Mean land surface temperature	Coefficients and odds ratios
(Conley *et al.* 2014)	RVF	*Cx. pipiens*	Larvae, adult	MaxEnt	Bioclim (mean diurnal range, maximum temperature of warmest	Not given

					month, minimum temperature of coldest month, annual temperature range, mean temperature of wettest/direst/warmest quarter)	
(Conley *et al.* 2014)	RVF	*Cx. pipiens*	Larvae, adult	Boosted regression trees	Bioclim (mean diurnal range, maximum temperature of warmest month, minimum temperature of coldest month, annual temperature range, mean temperature of wettest/direst/warmest quarter)	Not given
(Mweya *et al.* 2013)	RVF	*Cx. pipiens*	Adult	MaxEnt	Bioclim (isothermality, annual temperature range, mean temperature of warmest quarter, mean temperature of coolest quarter)	Not given
(Mweya *et al.* 2017)	RVF	*Cx. pipiens*	Adult	MaxEnt	Bioclim (mean temperature of warmest quarter, isothermality)	Not given
(Sallam *et al.* 2013)	RVF	*Cx. tritaeniorhynchus*	Larvae	MaxEnt	Bioclim (mean temperature of wettest quarter, mean temperature of driest quarter, mean diurnal range)	Not given
(Sánchez-Vizcaíno *et al.* 2013)	RVF	Human	NA	Multiple criteria decision making with weighted linear combination	Monthly mean temperature	Suitability and temperature
(Gleiser & Gorla 2007)	WEE	*Oc. albifasciatus*	Adult	Discriminant function analysis	Land surface temperature	Standardized coefficients
(Conley *et al.* 2014)	WNV	*Cx. pipiens*	Larvae, adult	MaxEnt	Bioclim (mean diurnal range, maximum temperature of warmest month, minimum temperature of coldest month, annual temperature range, mean temperature of wettest/direst/warmest quarter)	Not given
(Conley *et al.* 2014)	WNV	*Cx. pipiens*	Larvae, adult	Boosted regression trees	Bioclim (mean diurnal range, maximum temperature of warmest month, minimum temperature of coldest month, annual temperature range, mean temperature of wettest/direst/warmest quarter)	Not given

(Mongoh *et al.* 2007)	WNV	Horse	NA	Principle component analysis	Mean temperature	Estimates
(Tran *et al.* 2014)	WNV	Human	NA	Logistic regression model	Temperature anomalies	Parameters
(Ward 2005)	WNV	Horse	NA	Probability distribution model	Daily maximum temperature	Cumulative probabilities
(Ward *et al.* 2009)	WNV	Horse	NA	Bayesian spatial lag model	Mean temperature	Coefficient
(Messina *et al.* 2016)	ZIKV	*Aedes aegypti, Aedes albopictus*	NA	Boosted regression trees	WorldClim restricted by DENV transmission suitability	Not given
(Samy *et al.* 2016)	ZIKV	*Aedes aegypti, Aedes albopictus*	NA	MaxEnt	Land surface temperature	Not given
(Beebe *et al.* 2009)		*Aedes aegypti*	NA	GARP	Annual mean temperature, mean maximum and minimum temperatures for months of Jan and July	Not given
(Benedict *et al.* 2007)		*Aedes albopictus*	NA	GARP	Annual mean temperature, annual mean maximum temperature, annual mean minimum temperature, average DTR	Annual values for climatic environmental variables reported for georeferenced points
(Bogoch *et al.* 2016)		*Aedes aegypti, Aedes albopictus*	Eggs, larvae, pupae, adults	Boosted regression trees	Masked temperature suitability determined by dynamic population-level simulation	Not given
(Campbell *et al.* 2015)		*Aedes aegypti, Aedes albopictus*	NA	MaxEnt	WorldClim (monthly average maximum temperature, monthly average minimum temperature)	Not given
(Cardoso-Leite *et al.* 2014)		*Aedes aegypti*	NA	MaxEnt	Bioclim (annual mean temperature, mean diurnal range, isothermality, temperature seasonality, maximum temperature of warmest month, minimum temperature of coldest month, temperature annual range, mean temperature of wettest quarter, mean temperature of driest quarter, mean temperature of warmest quarter, mean temperature of coldest quarter)	Not given
(Cunze *et al.* 2016)		*Aedes albopictus, Aedes japonicus*	NA	MaxEnt and ensemble forecasting	WorldClim (mean temperature of coldest quarter, mean temperature	Not given

				of warmest quarter, temperature annual range)	
(Escobar *et al.* 2016b)	*Aedes aegypti, Aedes albopictus*	NA	MaxEnt	Land surface temperature	Not given
(Koch *et al.* 2016)	*Aedes albopictus*	NA	MaxEnt	WorldClim (annual mean temperature, mean temperature of warmest quarter, mean temperature of coldest quarter)	One variable response curves of modeled habitat suitability and temperate measures
(Kraemer *et al.* 2015)	*Aedes aegypti, Aedes albopictus*	Eggs, larvae, pupae, adults	Boosted regression trees	Temperature suitability based on previously published index	Not given
(Ogden *et al.* 2014)	*Aedes albopictus*	NA	Scaled suitability and sigmoidal function	Climatological indicators (thresholds for annual temperature, January temperature, summer temperature)	Not given
(Rochlin *et al.* 2013)	*Aedes albopictus*	Adult	MaxEnt	WorldClim (mean temperature of coldest quarter)	Null presence probability below -2°C mean winter temperature
(Shabani *et al.* 2018)	*Aedes albopictus*	Eggs, larvae, pupae, adults	MaxEnt	Bioclim (annual mean temperature of coldest quarter)	Highest probabilities of presence in areas where coldest mean quarter temperature ranged from 16-23°C

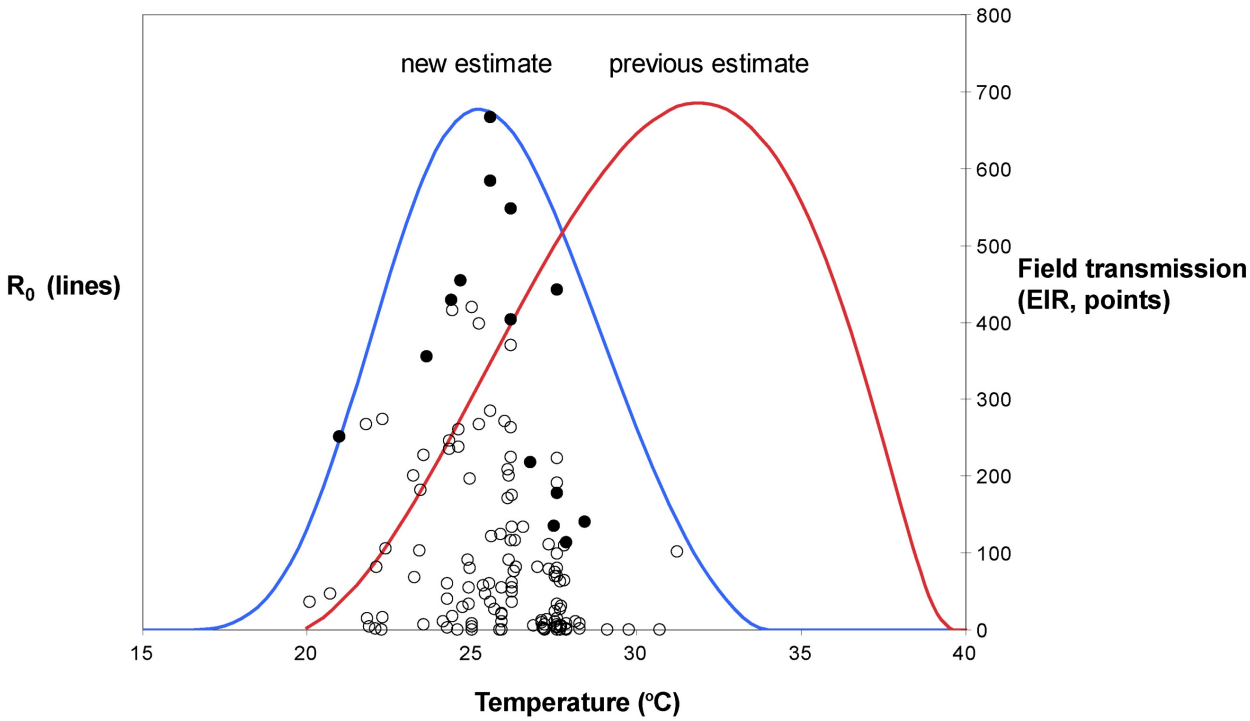

Figure S1. Validation of our malaria temperature-dependent R_0 model (blue line; Mordecai *et al.* 2013) in comparison with a previous model (red line; Parham & Michael 2010). Data points are independent field data on entomological inoculation rate (EIR) across Africa spanning 30 years (Hay *et al.* 2000) plotted against average temperature during the transmission season. Filled circles are the maximum EIR within each group of data points, binned by temperature (where we expect malaria to be the least controlled by other factors and therefore most sensitive to temperature). Open circles are the remaining data points. Reproduced with permission from (Mordecai *et al.* 2013)

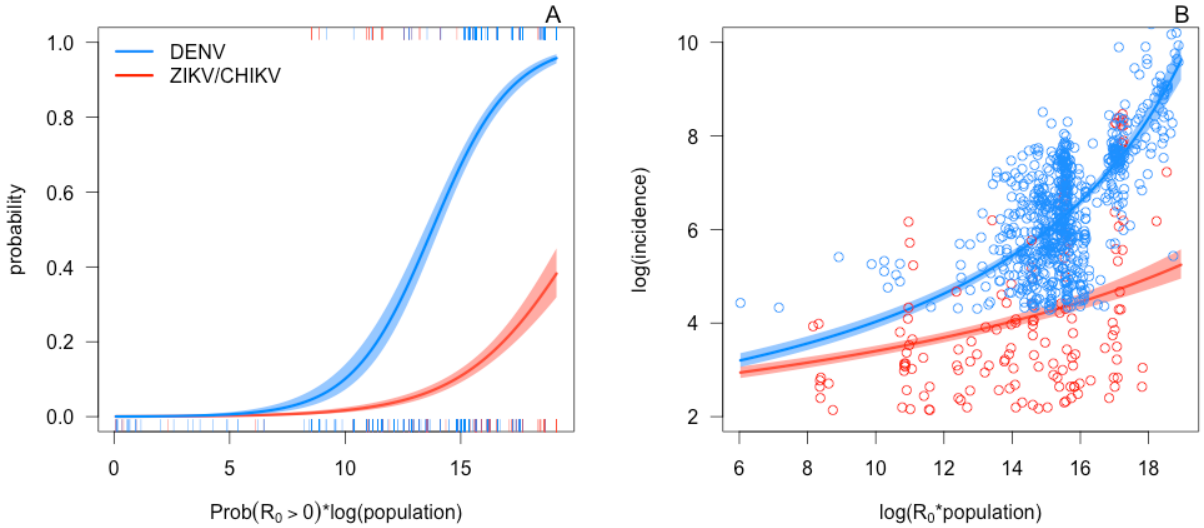

Fig. S2. Validation of our dengue, chikungunya, and Zika virus temperature-dependent R_0 model in *Aedes aegypti* (Mordecai *et al.* 2017). Lines compare the model-predicted dengue (blue) and chikungunya and Zika (red) probability occurrence of local transmission (A) and natural log of the incidence given that local transmission occurred (B) with weekly data country-scale data from the Americas in 2014-2016, plotted against the mean temperature 10 weeks prior (hashes in A and points in B). This 10-week lag represents our *a priori* expectation of the time that elapses between suitable temperatures and reported human cases (Stewart Ibarra *et al.* 2013; Mordecai *et al.* 2017). Predicted lines and 95% credible intervals (shaded areas) are the results of generalized linear models where the predictors are the posterior probability that $R_0(T) > 0$ times the log of population size (A), and the log of $R_0(T)$ times the population size (B). These metrics reflect our *a priori* expectation of the interactive relationship between R_0 and population size in driving epidemic occurrence and incidence. Reproduced with permission from (Mordecai *et al.* 2017).

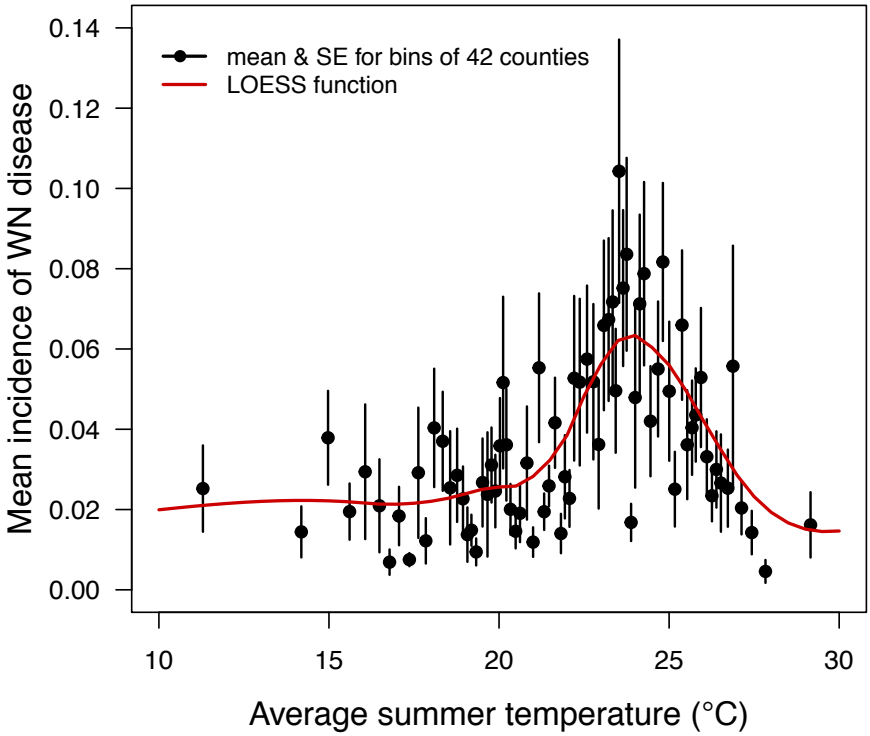

Fig. S3. Validation of our West Nile virus temperature-dependent R_0 model (Shocket et al., in prep). Points are incidence of human neuroinvasive West Nile disease across US counties, plotted against mean temperature from May – September from 2001-2016 for bins of 42 counties (+/- standard errors; total N = 3,109). Line is a LOESS regression illustrating the unimodal relationship between temperature and disease incidence with a peak at 24°C. Reproduced with permission from (Shocket et al., in prep).

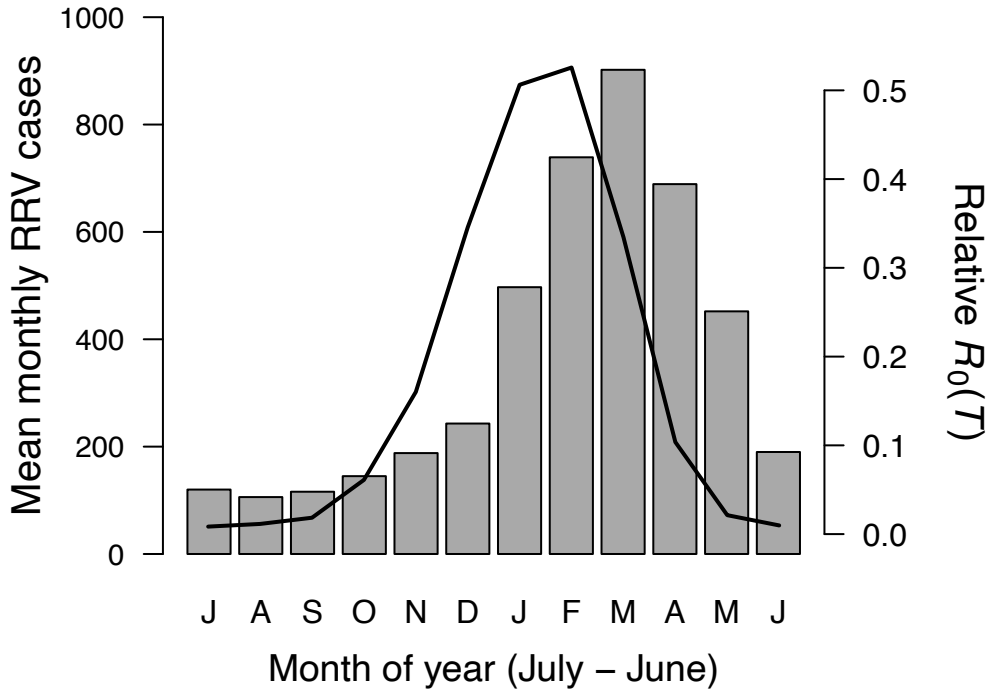

Fig. S4. Validation of our Ross River virus temperature-dependent R_0 model (Shocket *et al.* 2018). Average monthly human Ross River virus cases in Australia from 1992-2013 (gray bars) and average R_0 (line) plotted against average monthly temperature. Averages for temperature and R_0 are calculated across the 15 largest cities in Australia, weighted by population size. R_0 predicts the seasonality of Ross River virus incidence with a 2-month lag, matching our a priori expectation of the time that elapses between suitable temperatures and reported human cases (Stewart Ibarra *et al.* 2013; Mordecai *et al.* 2017). Reproduced with permission from (Shocket *et al.* 2018).

References Cited in Supplementary Materials

Alegana, V.A., Wright, J.A., Nahzat, S.M., Butt, W., Sediqi, A.W., Habib, N., *et al.* (2014). Modelling the Incidence of Plasmodium vivax and Plasmodium falciparum Malaria in Afghanistan 2006–2009. *PLoS ONE*, 9, e102304.

Arsevska, E., Hellal, J., Mejri, S., Hammami, S., Marianneau, P., Calavas, D., *et al.* (2016). Identifying Areas Suitable for the Occurrence of Rift Valley Fever in North Africa: Implications for Surveillance. *Transbound. Emerg. Dis.*, 63, 658–674.

Beebe, N.W., Cooper, R.D., Mottram, P. & Sweeney, A.W. (2009). Australia's Dengue Risk Driven by Human Adaptation to Climate Change. *PLoS Negl. Trop. Dis.*, 3, e429.

Benedict, M.Q., Levine, R.S., Hawley, W.A. & Lounibos, L.P. (2007). Spread of The Tiger: Global Risk of Invasion by The Mosquito Aedes albopictus. *Vector-Borne Zoonotic Dis.*, 7, 76–85.

Bhatt, S., Gething, P.W., Brady, O.J., Messina, J.P., Farlow, A.W., Moyes, C.L., *et al.* (2013). The global distribution and burden of dengue. *Nature*, 496, 504–507.

Bishop, A.L., Barchia, I.M. & Spohr, L.J. (2000). Models for the dispersal in Australia of the arbovirus vector, Culicoides brevitarsis Kieffer (Diptera: Ceratopogonidae). *Prev. Vet. Med.*, 47, 243–254.

Bogoch, I.I., Brady, O.J., Kraemer, M.U.G., German, M., Creatore, M.I., Kulkarni, M.A., *et al.* (2016). Anticipating the international spread of Zika virus from Brazil. *The Lancet*, 387, 335–336.

Calvete, C., Estrada, R., Miranda, M.A., Borrás, D., Calvo, J.H. & Lucientes, J. (2009). Ecological correlates of bluetongue virus in Spain: Predicted spatial occurrence and its relationship with the observed abundance of the potential Culicoides spp. vector. *Vet. J.*, 182, 235–243.

Campbell, L.P., Luther, C., Moo-Llanes, D., Ramsey, J.M., Danis-Lozano, R. & Peterson, A.T. (2015). Climate change influences on global distributions of dengue and chikungunya virus vectors. *Philos. Trans. R. Soc. B Biol. Sci.*, 370, 20140135.

Cappai, S., Loi, F., Coccollone, A., Contu, M., Capece, P., Fiori, M., *et al.* (2018). Retrospective analysis of Bluetongue farm risk profile definition, based on biology, farm management practices and climatic data. *Prev. Vet. Med.*, 155, 75–85.

Cardoso-Leite, R., Vilarinho, A.C., Novaes, M.C., Tonetto, A.F., Vilardi, G.C. & Guillermo-Ferreira, R. (2014). Recent and future environmental suitability to dengue fever in Brazil using species distribution model. *Trans. R. Soc. Trop. Med. Hyg.*, 108, 99–104.

Clements, A.C., Pfeiffer, D.U. & Martin, V. (2006). Application of knowledge-driven spatial modelling approaches and uncertainty management to a study of Rift Valley fever in Africa. *Int. J. Health Geogr.*, 5, 57.

Clements, A.C.A., Pfeiffer, D.U., Martin, V., Pittiglio, C., Best, N. & Thiongane, Y. (2007). Spatial Risk Assessment of Rift Valley Fever in Senegal. *Vector-Borne Zoonotic Dis.*, 7, 203–216.

Conley, A.K., Fuller, D.O., Haddad, N., Hassan, A.N., Gad, A.M. & Beier, J.C. (2014). Modeling the distribution of the West Nile and Rift Valley Fever vector Culex pipiens in arid and semi-arid regions of the Middle East and North Africa. *Parasit. Vectors*, 7, 289.

Conte, A., Ippoliti, C., Calistri, P., Pelini, S., Savini, L., Salini, R., *et al.* (2004). Towards the identification of potential infectious sites for bluetongue in Italy: a spatial analysis approach based on the distribution of Culicoides imicola. *Vet. Ital.*, 40, 311–315.

Cunze, S., Koch, L.K., Kochmann, J. & Klimpel, S. (2016). Aedes albopictus and Aedes japonicus - two invasive mosquito species with different temperature niches in Europe. *Parasit. Vectors*, 9, 573.

Escobar, L.E., Qiao, H. & Peterson, A.T. (2016a). Forecasting Chikungunya spread in the Americas via data-driven empirical approaches. *Parasit. Vectors*, 9, 112.

Escobar, L.E., Romero-Alvarez, D., Leon, R., Lepe-Lopez, M.A., Craft, M.E., Borbor-Cordova, M.J., *et al.* (2016b). Declining Prevalence of Disease Vectors Under Climate Change. *Sci. Rep.*, 6, 39150.

Foley, D.H., Linton, Y.-M., Ruiz-Lopez, J.F., Conn, J.E., Sallum, M.A.M., Póvoa, M.M., *et al.* (2014). Geographic distribution, evolution, and disease importance of species within the Neotropical Anopheles albitarsis Group (Diptera, Culicidae). *J. Vector Ecol.*, 39, 168–181.

Gao, X., Qin, H., Xiao, J. & Wang, H. (2017). Meteorological conditions and land cover as predictors for the prevalence of Bluetongue virus in the Inner Mongolia Autonomous Region of Mainland China. *Prev. Vet. Med.*, 138, 88–93.

Gleiser, R.M. & Gorla, D.E. (2007). Predicting the spatial distribution of Ochlerotatus albifasciatus (Diptera: Culicidae) abundance with NOAA imagery. *Bull. Entomol. Res.*, 97, 607–612.

Hay, S.I., Rogers, D.J., Toomer, J.F. & Snow, R.W. (2000). Annual Plasmodium falciparum entomological inoculation rates (EIR) across Africa: literature survey, internet access and review. *Trans. R. Soc. Trop. Med. Hyg.*, 94, 113–127.

Hundessa, S., Williams, G., Li, S., Liu, D.L., Cao, W., Ren, H., *et al.* (2018). Projecting potential spatial and temporal changes in the distribution of Plasmodium vivax and Plasmodium falciparum malaria in China with climate change. *Sci. Total Environ.*, 627, 1285–1293.

Koch, L.K., Cunze, S., Werblow, A., Kochmann, J., Dörge, D.D., Mehlhorn, H., *et al.* (2016). Modeling the habitat suitability for the arbovirus vector Aedes albopictus (Diptera: Culicidae) in Germany. *Parasitol. Res.*, 115, 957–964.

Kraemer, M.U., Sinka, M.E., Duda, K.A., Mylne, A.Q., Shearer, F.M., Barker, C.M., *et al.* (2015). The global distribution of the arbovirus vectors Aedes aegypti and Ae. albopictus. *eLife*, 4, e08347.

Messina, J.P., Kraemer, M.U., Brady, O.J., Pigott, D.M., Shearer, F.M., Weiss, D.J., *et al.* (2016). Mapping global environmental suitability for Zika virus. *eLife*, 5.

Mongoh, M.N., Khaitsa, M.L. & Dyer, N.W. (2007). Environmental and ecological determinants of West Nile virus occurrence in horses in North Dakota, 2002. *Epidemiol. Infect.*, 135, 57.

Mordecai, E.A., Cohen, J.M., Evans, M.V., Gudapati, P., Johnson, L.R., Lippi, C.A., *et al.* (2017). Detecting the impact of temperature on transmission of Zika, dengue, and chikungunya using mechanistic models. *PLoS Negl. Trop. Dis.*, 11, e0005568.

Mordecai, E.A., Paaijmans, K.P., Johnson, L.R., Balzer, C.H., Ben-Horin, T., de Moor, E., *et al.* (2013). Optimal temperature for malaria transmission is dramatically lower than previously predicted. *Ecol. Lett.*, 16, 22–30.

Mukhopadhyay, E., Hazra, S., Saha, G.K. & Banerjee, D. (2017). Altitudinal variation and bio-climatic variables influencing the potential distribution of Culicoides orientalis Macfie, 1932, suspected vector of Bluetongue virus across the North Eastern Himalayan belt of Sikkim. *Acta Trop.*, 176, 402–411.

Mweya, C.N., Kimera, S.I., Kija, J.B. & Mboera, L.E.G. (2013). Predicting distribution of Aedes aegypti and Culex pipiens complex, potential vectors of Rift Valley fever virus in relation to disease epidemics in East Africa. *Infect. Ecol. Epidemiol.*, 3, 21748.

Mweya, C.N., Kimera, S.I., Stanley, G., Misinzo, G. & Mboera, L.E.G. (2016). Climate Change Influences Potential Distribution of Infected Aedes aegypti Co-Occurrence with Dengue Epidemics Risk Areas in Tanzania. *PLOS ONE*, 11, e0162649.

Mweya, C.N., Mboera, L.E.G. & Kimera, S.I. (2017). Climate Influence on Emerging Risk Areas for Rift Valley Fever Epidemics in Tanzania. *Am. J. Trop. Med. Hyg.*, 97, 109–114.

Nsoesie, E., Kraemer, M., Golding, N., Pigott, D., Brady, O., Moyes, C., *et al.* (2016). Global distribution and environmental suitability for chikungunya virus, 1952 to 2015. *Euro Surveill. Bull. Eur. Sur Mal. Transm. Eur. Commun. Dis. Bull.*, 21.

Ogden, N.H., Milka, R., Caminade, C. & Gachon, P. (2014). Recent and projected future climatic suitability of North America for the Asian tiger mosquito Aedes albopictus. *Parasit. Vectors*, 7, 532.

Parham, P. & Michael, E. (2010). Modeling the Effects of Weather and Climate Change on Malaria Transmission. *Environ. Health Perspect.*, 118, 620–626.

Purse, B.V., Caracappa, S., Marino, A.M.F., Tatem, A.J., Rogers, D.J., Mellor, P.S., *et al.* (2004). Modelling the distribution of outbreaks and Culicoides vectors in Sicily: towards predictive risk maps for Italy. *Vet. Ital.*, 40, 303–310.

Rigot, T., Conte, A., Goffredo, M., Ducheyne, E., Hendrickx, G. & Gilbert, M. (2012). Predicting the spatio-temporal distribution of Culicoides imicola in Sardinia using a discrete-time population model. *Parasit. Vectors*, 5, 270.

Rochlin, I., Ninivaggi, D.V., Hutchinson, M.L. & Farajollahi, A. (2013). Climate Change and Range Expansion of the Asian Tiger Mosquito (Aedes albopictus) in Northeastern USA: Implications for Public Health Practitioners. *PLoS One San Franc.*, 8, e60874.

Sallam, M.F., Ahmed, A.M.A., Abdel-Dayem, M.S. & Abdullah, M.A.R. (2013). Ecological Niche Modeling and Land Cover Risk Areas for Rift Valley Fever Vector, Culex tritaeniorhynchus Giles in Jazan, Saudi Arabia. *PLOS ONE*, 8, e65786.

Samy, A.M., Thomas, S.M., Wahed, A.A.E., Cohoon, K.P., Peterson, A.T., Samy, A.M., *et al.* (2016). Mapping the global geographic potential of Zika virus spread. *Mem. Inst. Oswaldo Cruz*, 111, 559–560.

Sánchez-Vizcaíno, F., Martínez-López, B. & Sánchez-Vizcaíno, J.M. (2013). Identification of suitable areas for the occurrence of Rift Valley fever outbreaks in Spain using a multiple criteria decision framework. *Vet. Microbiol.*, 165, 71–78.

Shabani, F., Tehrany, M.S., Solhjouy-fard, S. & Kumar, L. (2018). A comparative modeling study on non-climatic and climatic risk assessment on Asian Tiger Mosquito (Aedes albopictus). *PeerJ*, 6, e4474.

Shocket, M.S., Ryan, S.J. & Mordecai, E.A. (2018). Temperature explains broad patterns of Ross River virus transmission. *eLife*, 7, e37762.

Stewart Ibarra, A.M., Ryan, S.J., Beltrán, E., Mejía, R., Silva, M. & Muñoz, Á. (2013). Dengue vector dynamics (Aedes aegypti) influenced by climate and social factors in Ecuador: implications for targeted control. *PLoS ONE*, 8, e78263.

Tatem, A.J., Baylis, M., Mellor, P.S., Purse, B.V., Capela, R., Pena, I., *et al.* (2003). Prediction of bluetongue vector distribution in Europe and north Africa using satellite imagery. *Vet. Microbiol.*, 97, 13–29.

Tran, A., Sudre, B., Paz, S., Rossi, M., Desbrosse, A., Chevalier, V., *et al.* (2014). Environmental predictors of West Nile fever risk in Europe. *Int. J. Health Geogr.*, 13, 26.

Ward, M.P. (2005). Epidemic West Nile virus encephalomyelitis: A temperature-dependent, spatial model of disease dynamics. *Prev. Vet. Med.*, 71, 253–264.

Ward, M.P., Wittich, C.A., Fosgate, G. & Srinivasan, R. (2009). Environmental risk factors for equine West Nile virus disease cases in Texas. *Vet. Res. Commun.*, 33, 461–471.